DA5400W (Foundations of Machine Learning) Assignment 1

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Problem 1

The following probability distribution function can describe the growth rate of finance:

$$f(x) = \frac{1}{\alpha^2} x e^{-x/\alpha}$$

with $\alpha \in (0, \infty)$ and $x \in [0, \infty]$. Find an estimate of the parameter α using the maximum likelihood estimators and the method of moments given the datasets $x_1, x_2, \cdots x_n$.

Solution

Sample Size is denoted as n here.

- Estimating using **Maximum Likelihood Estimator** :

Likelihood is joint probability of sample, i.e. product of probabilites of each input x_i (given parameter α):

$$L(\alpha; \mathbf{x}) = \Pi_{i=1}^{N} f(x_i, \alpha) = \Pi_{i=1}^{N} \frac{1}{\alpha^2} x_i e^{-x_i/\alpha} = \frac{1}{\alpha^{2N}} \Pi_{i=1}^{n} x_i e^{-x_i/\alpha}$$

We need to maximize likelihood, or equivalently maximize log-likelihood:

$$l(\alpha; \mathbf{x}) = -2n ln(\alpha) + \sum_{i=1}^{n} ln(x_i) - \sum_{i=1}^{n} \frac{x_i}{\alpha}$$

Solving optimization problem $\max_\alpha l(\alpha; \mathbf{x})$:

$$\frac{d}{d\alpha} l(\alpha; \mathbf{x}) = 0 \implies \frac{-2n}{\alpha} + 0 + \frac{1}{\alpha^2} \sum_{i=1}^{n} x_i = 0 \implies \alpha = \frac{1}{2n} \sum_{i=1}^{n} x_i$$
$$\frac{d^2}{d\alpha^2} l(\alpha; \mathbf{x}) = \frac{2}{\alpha^2} + \frac{2}{\alpha^3} > 0 \implies \text{Stationary point is maxima}$$

- Estimating using **Method of Moments**:

As there is only 1 parameter, so only 1st moment of theoritical and sample mean needs to be equated:

$$E[X \mid \theta] = E[x]$$
$$\implies \int_0^\infty x f(x) dx = \frac{1}{n} \sum_{i=1}^{n} x_i$$
$$\implies \int_0^\infty x \frac{1}{\alpha^2} x e^{-x/\alpha} dx = \frac{1}{n} \sum_{i=1}^{n} x_i$$
$$\implies \frac{1}{\alpha^2} \int_0^\infty x^2 e^{-x/\alpha} dx = \frac{1}{n} \sum_{i=1}^{n} x_i$$
$$\implies \frac{1}{\alpha^2} 2! \alpha^3 = \frac{1}{n} \sum_{i=1}^{n} x_i \quad \text{(using Gaussian integral:} \int_0^\infty x^2 e^{-x/\alpha} = 2!\alpha^3)$$
$$\implies \alpha = \frac{1}{2n} \sum_{i=1}^{n} x_i$$

Therefore both methods give same estimated parameter.

Problem 2

Consider a process for which a random sample $X_1, X_2, \cdots X_9$ is collected to understand the population properties. The population mean and variance are μ and σ^2. Mr and Mrs. Stat have proposed two estimators as follows:

$$\theta_{Mr} = \frac{X_1 + X_3 + 4X_5}{6}, \quad \theta_{Mrs} = \frac{X_2 - X_6 + 2X_7 + 3X_4}{5}$$

Which is the best estimator and why?

Solution

Using $E(aX \pm bY) = aE(X) \pm aE(Y), Var(aX \pm bY) = a^2 Var(X) + b^2 Var(Y)$:

$$E[\theta_{Mr}] = \frac{\mu + \mu + 4\mu}{6} = \mu, \quad Var\theta_{Mr} = \frac{\sigma^2 + \sigma^2 + 4^2\sigma^2}{6^2} = 0.5\sigma^2$$
$$E[\theta_{Mrs}] = \frac{\mu - \mu + 2\mu + 3\mu}{5} = \mu, \quad Var(\theta_{Mrs}) = \frac{\sigma^2 + \sigma^2 + 2^2\sigma^2 + 3^2\sigma^2}{5^2} = 0.6\sigma^2$$

Both estimators have expected value μ so are **unbiased**, but θ_{Mr} has lower variance so it's better estimator.

Problem 4

Find the maximum likelihood estimate of the parameter θ of the following probability distribution function:

$$f(y;\theta) = \frac{3y^2}{\theta^3}$$

with $\theta \in (0,\infty)$ and $y \in [0,\theta]$ using the data $y_1, y_2, \cdots, y_n$.

Solution

Likelihood is: $L(\theta; y) = \Pi_{i=1}^{n} \frac{3y_i^2}{\theta^3} = \frac{3^n}{\theta^3} \Pi_{i=1}^{n} y_i^2$

We need to find $\max_\theta l(\theta; y) = \max_\theta \frac{3^n}{\theta^3} \Pi_{i=1}^{n} y_i^2$.

θ is in denominator so we want to minimize it. But $y \in [0,\theta]$ so minimum value it can take is max input $\theta = \max(y_1, y_2, \cdots, y_n)$.

Problem 6

Consider a random variable C with mean μ and standard deviation σ. Two experiments are performed to collect two random samples with n_1 and n_2 sample sizes. The sample means for these experiments are C_1 and C_2 . Then, show that an estimator for the sample mean:

$$\bar{C} = a\bar{C}_1 + (1-a)\bar{C}_2$$

is an unbiased estimator for the mean μ.

Solution

$E[a\bar{C}_1 + (1-a)\bar{C}_2] = a\mu + (1-a)\mu = \mu = \bar{C}$: So the given estimator for mean is unbiased, as expected value of estimator equals mean.

Problem 7

Find the local extrema of the following functions and classify the points as minimum or maximum:

1. $f(x) = 4x^3 - 3x^2 + 2x - 1$
2. $f(x) = sin(x) + cos(x)$
3. $f(x) = \frac{x^2-1}{x}$

Solution

1. $\nabla f = 12x^2 - 6x + 2 = 0$: No real solutions exist for x so no stationary points / extrema exist.
2.

$$\nabla f = cos(x) - sin(x) = 0 \implies x = \frac{sin^{-1}(1)}{2}$$
$$\nabla^2 f = -sin(x) - cos(x) \implies \nabla^2 f(\frac{sin^{-1}(1)}{2}) = -\sqrt{1 + sin(2x)} = -\sqrt{2}$$

So one extrema exists: $\frac{sin^{-1}(1)}{2}$ (maxima since $\nabla^2 f > 0$).

3. $f(x) = x - x^{-1}, \quad \nabla f = 1 + x^{-2} = 0$: No real solutions exist for x so no stationary points / extrema exist.

Problem 8

Show that the function $f(x_1, x_2) = 8x_1 + 12x_2 + x_1^2 - 2x_2^2$ has only one stationary point and that it is neither a maximum nor a minimum, but a saddle point. Sketch the contour line of f.

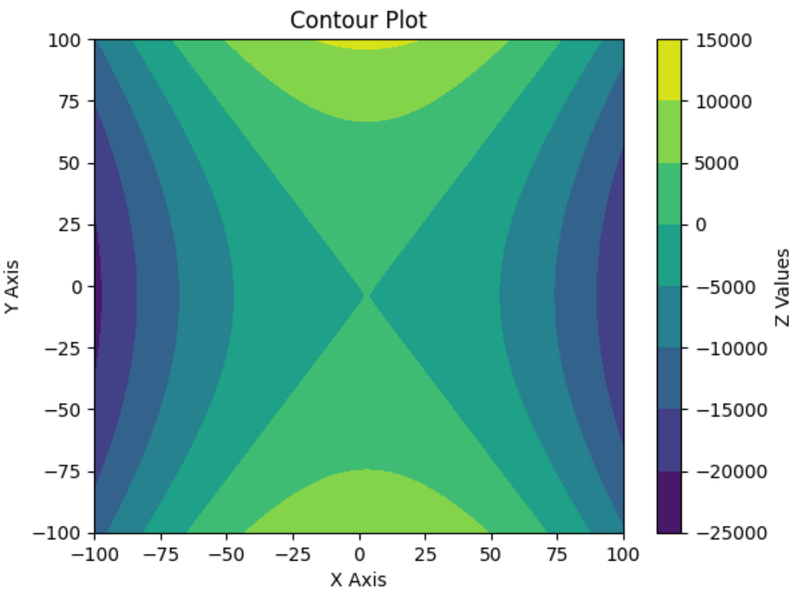
Solution

$$\nabla f = \begin{pmatrix} 8 + 2x_1 \\ 12 - 4x_2 \end{pmatrix} = 0 \implies (x_1, x_2) = (-4, 3)$$
$$\nabla^2 f = \begin{pmatrix} 2 & 0 \\ 0 & -4 \end{pmatrix}$$

Hessian $\nabla^2 f$ is a diagonal matrix so eigenvalues are from diagonal: 2, -4. As one positive and one negative eigenvalue is there, it's an indeterminate matrix. So stationary point $(-4, 3)$ is a saddle point.

Plotting contour (2D visualization of a 3D surface):

```python
import numpy as np
from matplotlib import pyplot as plt
x = np.linspace(-100,100,201)
y = np.linspace(-100,100,201)
X, Y = np.meshgrid(x, y, indexing='ij')
Z = 8*X + 12*Y + X**2 - 2*Y**2
plt.contourf(x,y,Z)
plt.colorbar(label='Z Values')
plt.title('Contour Plot')
plt.xlabel('X Axis')
plt.ylabel('Y Axis')
plt.show()
```



Problem 9

Determine the stationary points and classify their nature for the function $f(x, y) = x^4 + y^4 - 36xy$.

Solution

$$\nabla f = \begin{pmatrix} 4x^3 - 36y \\ 4y^3 - 36x \end{pmatrix} = 0 \implies 4x^3 = 30y, 4y^3 = 30x \implies (x, y) = (0,0), (3,3), (-3,-3)$$

$$\nabla^2 f = \begin{pmatrix} 12x^2 & -36 \\ -36 & 12y^2 \end{pmatrix}$$

$$\nabla^2 f(0,0) = \begin{pmatrix} 0 & -36 \\ -36 & 0 \end{pmatrix} \implies det(\nabla^2 f(0,0)) = -36^2 < 0 \implies \text{Saddle Point} \nabla^2 f(\pm 3, \pm 3) = \begin{pmatrix} 108 & -36 \\ -36 & 108 \end{pmatrix} \implies det(\nabla^2 f(\pm 3, \pm 3)) = 108^2 - 36^2 > 0, f_{xx} = 108$$

So stationary points are (0,0) (saddle point) and (3,3), (-3,-3) (both local minima).

This works because for 2x2 Hessian Matrix:

- determinant $D < 0$ implies eigen values λ_1, λ_2 are of opposite signs (as $D = \lambda_1 \lambda_2$), so point is saddle point.
- determinant $D > 0$ and $f_{xx} > 0$ implies positive eigen values, so Hessian is Positive Definite and point is local minima.

Problem 10

Solve the following optimization problems by hand(s) and also draw the feasible regions:

1. Find the maximum of the following functions:
   - $f(x) = 1 - 8x + 2x^2 - \frac{10}{3}x^3 + \frac{1}{4}x^4 + \frac{4}{5}x^5 - \frac{1}{6}x^6$
   - $f(x_1, x_2) = x_1 + x_2 \quad \text{subject to} \quad x_1^2 + x_2^2 - 1 = 0$
2. Verify the KKT conditions and find the Lagrange multipliers for the following function at $x = (1, 0)$:

$$(x_1 - \frac{3}{2})^2 + (x_2 - \frac{1}{8})^2$$

$$\text{subject to} \quad \begin{pmatrix} 1 - x_1 - x_2 \\ 1 - x_1 + x_2 \\ 1 + x_1 - x_2 \\ 1 + x_1 + x_2 \end{pmatrix} \geq 0$$

3. Find the minimum of the function $f(x) = (x_1 - 1)^2 + x_2^2, \quad \text{subject to} \quad x_1 - x_2^2 \leq 0$

Solution

1.

- $f(x) = 1 - 8x + 2x^2 - \frac{10}{3}x^3 + \frac{1}{4}x^4 + \frac{4}{5}x^5 - \frac{1}{6}x^6$:

$$\nabla f = -8 + 4x - 10x^2 + x^3 + 4x^4 - x^5 = (-x^5 + 2x^4 + 5x^3 + 4x) + (2x^4 - 4x^3 - 10x^2 - 8) = (x-2)(-x^4 + 2x^3 + 5x^2 + 4) = 0$$
$$\nabla^2 f = 4 - 20x + 16x^3 - 5x^4$$

Here one root (stationary point) is 2; remaining real roots are approximately 3.51471704 and -1.71339702 (found using `numpy.polynomial.Polynomial([4,0,5,2,-1]).roots()`).

For each of these roots:

$$\nabla^2 f(2) = 4 - 20 * 2 + 16 * 2^3 - 5 * 2^4 = 12 > 0 \quad \text{(local minima)}$$
$$\nabla^2 f(3.51471704) = 4 - 20 * 3.51471704 + 16 * 3.51471704^3 - 5 * 3.51471704^4 = -134.6 < 0 \quad \text{(local maxima)}$$
$$\nabla^2 f(-1.71339702) = 4 - 20 * (-1.71339702) + 16 * (-1.71339702)^3 - 5 * (-1.71339702)^4 = -85.3 < 0 \quad \text{(local maxima)}$$

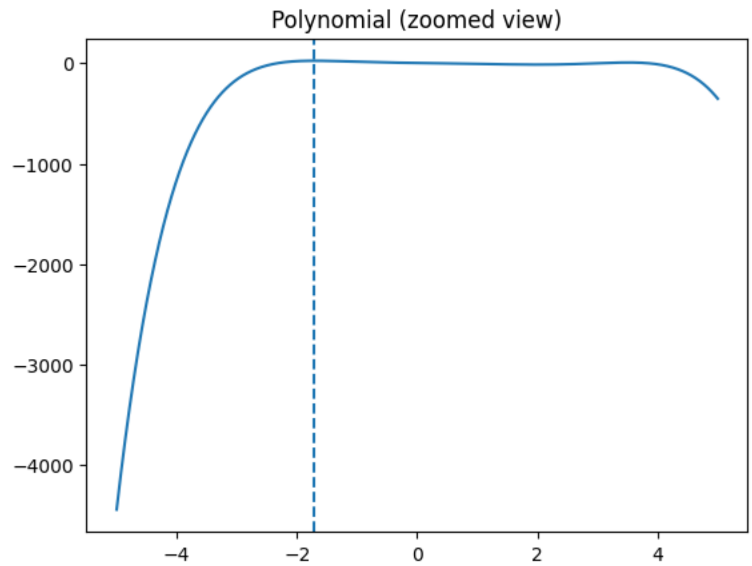
Since 2 local maxima exist, find which has greater value:

$$f(3.51471704) = 1 - 8 * 3.51471704 + 2 * 3.51471704^2 - 3.51471704^3 * 10/3 + 3.51471704^4/4 + 3.51471704^5 * 4/5 - 3.51471704^6/6 = 5.906$$
$$f(-1.71339702) = 1 - 8 * (-1.71339702) + 2 * (-1.71339702)^2 - (-1.71339702)^3 * 10/3 + (-1.71339702)^4/4 + (-1.71339702)^5 * 4/5 - (-1.713397$$

So global maxima is -1.71339702 and maximum value of $f(x)$ is 23.469 .

Plotting feasible region:

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-5, 5, 2000)
y = 1 - x*8 + x**2*2 - x**3*10/3 + x**4/4 + x**5*4/5 - x**6/6
plt.plot(x, y)
plt.axvline(-1.71339702, linestyle="--")    # add dotted line showing maxima point
plt.title("Polynomial (zoomed view)")
plt.show()
```



- $f(x_1, x_2) = x_1 + x_2 \quad \text{subject to} \quad x_1^2 + x_2^2 - 1 = 0$

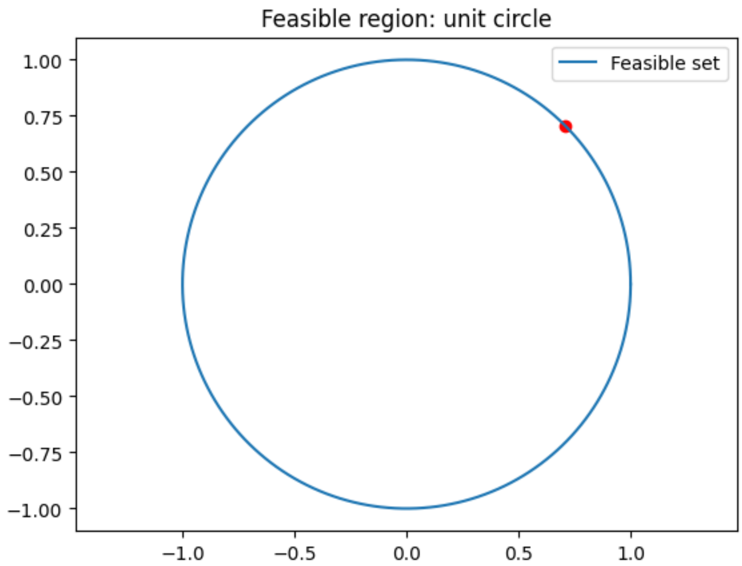
(x_1, x_2) lies on a circle of radius 1 around origin. Let θ be angle of point wrt X axis such that $x = cos(\theta), y = sin(\theta)$. Then:

$$f(\theta) = cos(\theta) + sin(\theta)$$
$$\nabla_\theta f = -sin(\theta) + cos(\theta) = 0 \implies \theta = \frac{pi}{4}$$
$$\nabla^2_\theta f = -cos(\theta) - sin(\theta) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2} < 0 \quad \text{(maxima)}$$

So maxima point is $(x_1, x_2) = (cos(\frac{\pi}{4}), sin(\frac{\pi}{4})) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and maximum value is $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$.

Plotting feasible region:

```python
import numpy as np
from matplotlib import pyplot as plt
theta = np.linspace(0, 2*np.pi, 400)
x1 = np.cos(theta)
x2 = np.sin(theta)
plt.plot(x1, x2, label="Feasible set")
plt.scatter([1/np.sqrt(2)], [1/np.sqrt(2)], color="red")
plt.axis("equal")
plt.legend()
plt.title("Feasible region: unit circle")
plt.show()
```



2.

$$
\begin{aligned}
& (x_1 - \frac{3}{2})^2 + (x_2 - \frac{1}{8})^2 \\
\text{subject to} & \begin{pmatrix} 1 - x_1 - x_2 \\ 1 - x_1 + x_2 \\ 1 + x_1 - x_2 \\ 1 + x_1 + x_2 \end{pmatrix} \geq 0
\end{aligned}
$$

Lagrangian: $L(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (x_1 - \frac{3}{2})^2 + (x_2 - \frac{1}{8})^2 + \lambda_1(x_1 + x_2 - 1) + \lambda_2(x_1 - x_2 - 1) + \lambda_3(-x_1 + x_2 - 1) + \lambda_4(-x_1 - x_2 - 1)$ where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are Lagrangian multipliers.

KKT Conditions are:

- Stationarity:

$$\nabla_{x_1, x_2} L = \binom{2x_1 - 3 + \lambda_1 + \lambda_2 - \lambda_3 - \lambda_4}{2x_2 - \frac{1}{4} + \lambda_1 - \lambda_2 + \lambda_3 - \lambda_4} = 0$$

- Primal Feasability:

$$\begin{pmatrix} x_1 + x_2 - 1 \\ x_1 - x_2 - 1 \\ -x_1 + x_2 - 1 \\ -x_1 - x_2 - 1 \end{pmatrix} \leq 0$$

- Dual Feasability:

$$\lambda_1, \lambda_2, \lambda_3, \lambda_4 \geq 0$$

- Complementary Slackness:

$$
\begin{aligned}
\lambda_1(x_1 + x_2 - 1) = 0 \\
\lambda_2(x_1 - x_2 - 1) = 0 \\
\lambda_3(-x_1 + x_2 - 1) = 0 \\
\lambda_4(-x_1 - x_2 - 1) = 0
\end{aligned}
$$

At $(x_1 = 1, x_2 = 0)$:

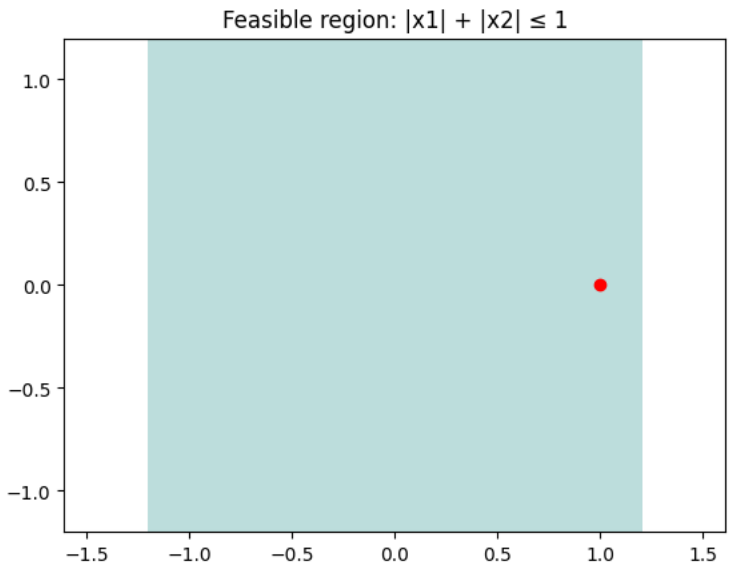
$$
\begin{aligned}
\lambda_1(1 + 0 - 1) = 0 &\implies 0\lambda_1 = 0 \\
\lambda_2(1 - 0 - 1) = 0 &\implies 0\lambda_2 = 0 \\
\lambda_3(-1 + 0 - 1) = 0 &\implies -2\lambda_3 = 0 \implies \lambda_3 = 0 \\
\lambda_4(-1 - 0 - 1) = 0 &\implies -2\lambda_4 = 0 \implies \lambda_4 = 0
\end{aligned}
$$

$$\begin{pmatrix} 2(1) - 3 + \lambda_1 + \lambda_2 - 0 - 0 \\ 2(0) - \frac{1}{4} + \lambda_1 - \lambda_2 + 0 - 0 \end{pmatrix} = 0$$
$$\implies \begin{pmatrix} \lambda_1 + \lambda_2 - 1 \\ \lambda_1 - \lambda_2 - \frac{1}{4} \end{pmatrix} = 0$$
$$\implies \lambda_1 = \frac{5}{8}, \lambda_2 = \frac{3}{8}$$

Therefore Lagrangian multipliers are $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (\frac{5}{8}, \frac{3}{8}, 0, 0)$

Plotting feasible region:

```
import numpy as np
from matplotlib import pyplot as plt
x = np.linspace(-1.2, 1.2, 400)
X, Y = np.meshgrid(x, x)
feasible = (np.abs(X) + np.abs(Y)) <= 1
plt.contourf(X, Y, feasible, levels=[0,1], alpha=0.3)
plt.scatter(1, 0, color="red")  # KKT point
plt.axis("equal")
plt.title("Feasible region: |x1| + |x2| ≤ 1")
plt.show()
```



3. $\min f(x) = (x_1 - 1)^2 + x_2^2, \quad \text{subject to} \quad x_1 - x_2^2 \leq 0$

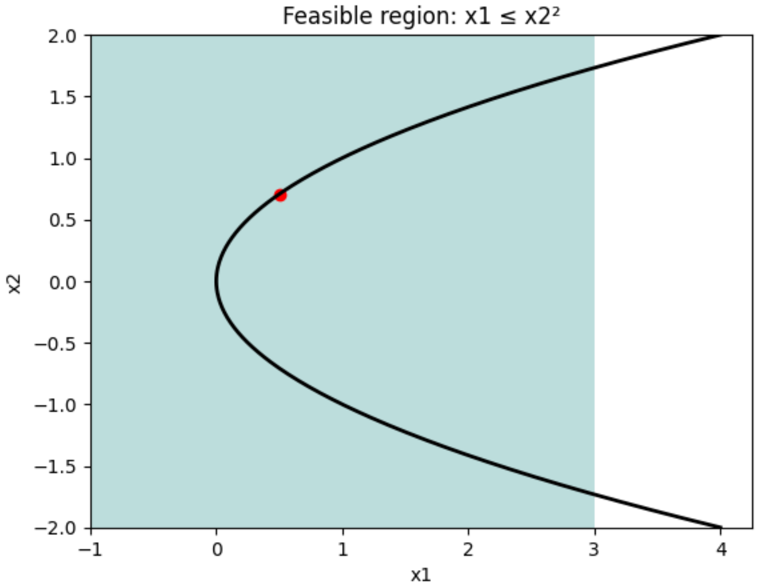
In terms of x_2:

$$f(x_2) = x_1^2 - 2x_1 + 1 + x_2^2 = x_2^4 - 2x_2^2 + 1 + x_2^2 = x_2^4 - x_2^2 + 1$$
$$f'(x_2) = 4x_2^3 - 2x_2 = 0 \implies x_2 = 0, \frac{1}{\sqrt{2}}$$
$$f''(x_2) = 12x_2^2 - 2 \implies f''(0) = -2 < 0, f''(\frac{1}{\sqrt{2}}) = 4 > 0$$

So minima is at $(x_1, x_2) = (\frac{1}{2}, \frac{1}{\sqrt{2}})$ and minimum value is $(\frac{1}{2} - 1)^2 + \frac{1}{\sqrt{2}}^2 = \frac{3}{4}$.

Plotting feasible region:

```
import numpy as np
from matplotlib import pyplot as plt
x2 = np.linspace(-2, 2, 400)
x1 = np.linspace(-1, 3, 400)
X1, X2 = np.meshgrid(x1, x2)
feasible = X1 <= X2**2
plt.contourf(X1, X2, feasible, levels=[0,1], alpha=0.3)
plt.plot(x2**2, x2, color="black", linewidth=2)
plt.scatter(0.5, 1/np.sqrt(2), color="red")  # minimizer
plt.xlabel("x1")
plt.ylabel("x2")
plt.title("Feasible region: x1 ≤ x2²")
plt.show()
```


Feasible region: x1 ≤ x2²
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
x2
−1
0
1
2
3
4
x1